

SFWRENG 3MX3 - Assignment 8

Matthew Farah, farahm11@mcmaster.ca, 400468251

December 5, 2025

Contents

Exercise 1

(a) Compute the transfer function of the system $y(n) = x(n) - y(n-1) + x(n-1)$.

Solution:

For a discrete system given in difference equation form $y(n) = \sum_{k=0}^M \alpha_k x(n-k) + \sum_{k=1}^N \beta_k y(n-k)$ its transfer function, $G(z)$, is given by $G(z) = \frac{\sum_{k=0}^M \alpha_k z^{-k}}{1 - \sum_{k=1}^N \beta_k z^{-k}} \hat{X}(z)$. Therefore, to find the transfer function, we must first compute $\hat{X}(z)$.

(I) Compute $\hat{X}(z)$.

(b) What is the frequency response of the system with the transfer function $G(z) = \frac{z}{.5-z}$?

Solution: